

Development, characterization, and parametric analysis of dry sliding wear behavior of epoxy-short betel nut fiber composite using response surface method and neural computation

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Purushottam Kumar Choudhary , Bishnu Prasad Nanda and Alok Satapathy

Abstract

This work reports on the mechanical and wear performance of epoxy composite reinforced with short betel nut fiber (SBF). Composite samples with different weight percentages (0, 2, 3, 4, 6, and 8 wt%) of fiber content are fabricated through hand lay-up route. Mechanical properties such as tensile and flexural strengths are evaluated by conducting tests as per appropriate ASTM standards. Sliding wear tests are performed on a pin-on-disc test apparatus as per ASTM G99 standard. A non-linear regression model is developed in accordance with face-centered central composite design (FCCCD) of Response Surface Methodology (RSM). An artificial neural network (ANN) approach is applied to predict the wear rate of the composite and compared with the RSM predicted results. It is found that with the incorporation of short betel nut fiber both tensile and flexural strength of the composite shows an increasing trend. It is also observed that reinforcement of short betel nut fiber enhances the wear performance of epoxy. Surface morphologies of the worn samples have been studied to analyze the wear mechanism of the composite samples.

Keywords

Betel nut fiber, epoxy, dry sliding wear, response surface methodology, Artificial neural network (ANN)

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Introduction

Recently, natural fiber-reinforced polymer composites have acquired considerable attention due to their various attractive features like low cost, easy availability, high toughness, biodegradability, and high recyclability. Nowadays a substantial growth has been witnessed in the use of natural fiber-reinforced composites in various fields of application like in construction, aerospace, biomedical, and marine. In the case of automobile sector, several portions of vehicles such as door panels, dashboards, inside parts, and the manufacturing of bearings used for gears are also produced by natural fiber polymer composites.^{1–5}

In the tropical and equatorial regions, fiber-based plants like coir, oil palm, banana, bamboo, and jute are found in abundant quantity. Fibers extracted from these plants are proved to be an appropriate and powerful reinforcement for various kinds of thermoset and thermoplastic polymers.^{6–8} It is worth noting that the tribological properties of polymers can be improved with the reinforcement with some specific natural fibers.⁹ Many research reports are available on the friction and wear performance of natural fiber-reinforced composite. Yallew et al.¹⁰ studied the wear behavior of polypropylene-hemp fiber composite and determined that the wear rate and coefficient of friction are reduced with the reinforcement of fiber. Patnaik et al.¹¹ investigated the sliding wear behavior of pine-bark fiber-reinforced polyester composite and concluded that wear behavior is highly influenced by the fiber content in the composite. Ahmed et al.¹² evaluated the sliding wear performance of jute/epoxy composite filled with SiC/Al₂O₃ fillers and found that the addition of these fillers slightly decrease the wear rate of the composite. It also reveals that Al₂O₃ filled composite has higher performance than SiC filled composite.

Department of Mechanical Engineering, National Institute of Technology, Rourkela, India

Corresponding author:

Purushottam Kumar Choudhary, Department of Mechanical Engineering, National Institute of Technology, Rourkela 769008, India.

Email: kumarpurushottam41@gmail.com



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Betel nut is a species of palm, which develops in high dampness content of soil. It is extensively found in East Africa, South Asia, and the Pacific islands. The outer layer of the fiber surface has trichomes and due to the occurrence of trichomes higher interfacial bonding can be obtained with the reinforcement of betel nut fiber in the polymer matrix.^{13–16} Many research reports are available on various characteristics of betel nut fiber-reinforced polymer composite. Hassan et al.¹⁷ examined the mechanical properties of jute/betel nut hybrid composite and concluded that due to alkalization, the performance of composites is improved. Rahman et al.¹⁸ investigated chemically treated hybrid coir/betel nut composite and observed that an equal proportion of fiber in the composite has better mechanical performance than composite containing fibers at a ratio of 3:1 or 1:3. However, in the authors' knowledge, no work has been reported so far on the sliding wear analysis of short betel nut fiber-reinforced polymer composites using response surface methodology (RSM) and artificial neural network (ANN). Thus in the present research, the sliding wear behavior of short betel nut fiber-reinforced epoxy composite has been analyzed using RSM, and ANN is used for the prediction of the specific wear rate of the composites.

Experimental details

Materials

Betel nut fruits are collected from the local market. Shells are extracted from nuts and cleaned with water to remove impurities. Then shells are dried in sunlight for 5 days to remove its moisture and smell. Fibers are extracted from the husk and again dried in a microwave at 70°C for 90 min to remove its remaining moisture content. Fibers are cut roughly into 4–5 mm sizes manually. Some important properties of betel nut fiber are presented in Table 1.¹⁹

Epoxy is a widely used matrix material due to its certain useful properties. In the present research, epoxy LY-556 and its corresponding hardener HY-951 are used as the matrix material. Some important properties of epoxy are presented in Table 2.

Composite fabrication

Fabrication of composite with epoxy as matrix and short betel nut fiber as reinforcing material is done by a simple hand lay-up process. To fabricate the composite epoxy is mixed with hardener at a ratio of 10:1 by weight and various proportions of fibers are then added into the mixture and stirred with a glass rod so that fibers are uniformly distributed in the mixture. The mixture is poured into a cylindrical and a rectangular mold and allowed to solidify at room temperature. For easy removal of the castings silicone releasing spray is sprayed before pouring the mixture into the mold and a reasonable amount of load is applied on the casting. The cylindrical composites samples are fabricated with 0, 3, and 6 wt% of fiber content for sliding wear test and the rectangular composite samples are prepared with 0, 2, 4, 6, and 8 wt% of fiber content for flexural and tensile tests.

Mechanical properties

The tensile and three-point bending tests in the present research are performed in an Instron-5967 UTM machine to obtain the tensile strength and flexural strength of the composite samples. A total of three samples of each fiber content are tested in both cases and their average value is reported as the tensile and flexural strength of the composite. The tensile tests are carried out as per ASTM D-638 standard with a crosshead speed of 10 mm/min and the three-point bending tests are performed in accordance with ASTM D-790 with a crosshead speed of 2 mm/min.

Flexural Strength is evaluated by²⁰

$$F.S. = \frac{3PL}{2bt^2} \quad (1)$$

Table 1. Properties of betel nut fiber.

Properties	Value
Density	0.38 g/cm ³
Tensile strength	128.79 MPa
Young's modulus	2569 MPa

Table 2. Properties of epoxy resin.

Properties	Value
Density	1.15 g/cm ³
Compressive strength	90.0 MPa
Micro-hardness	0.0850 GPa

where $F.S.$ is flexural strength (MPa), P is applied load (N), L is span length (m), and b and t is breadth (m) and thickness (m) of the test specimen, respectively.

Sliding wear test

A pin-on-disc test machine is used to investigate the dry sliding wear behavior of epoxy-SBF composites. The tests are conducted according to the ASTM G-99 standard. The specimen of size 4 mm radius and 30 mm length is used for the tests. The tests are conducted by fixing the emery paper of grit size P220 on the steel disc with a standard adhesive. The test specimen is then allowed to slide against the emery paper. A high accuracy digital electronic weighing machine (up to 0.001 mg accuracy) is used to measure the weight of the samples before and after performing the trial. The entire trial is conducted thrice to ensure consistency in test results. The different test parameters considered for the wear evaluation are sliding velocity, sliding distance, normal load, and fiber content.

Specific wear rate (W_s) of the composite is evaluated by⁵

$$W_s = \frac{\Delta m}{\rho \times t \times V_s \times F_n} \quad (2)$$

where Δm is mass loss of specimen (g), ρ is density of specimen (g/cm^3), t is period of the test (s), V_s is sliding velocity (cm/s), and F_n is normal load (N).

Response surface methodology

The response surface methodology (RSM) is a statistical technique generally used for modeling problems and to find a correlation between the response and the independent variables. In the present investigation, a face-centered central composite design (FCCCD) is considered for the analysis. FCCCD comprises a lot of axial points, factorial points, and center points. Axial points and factorial points are accountable for the estimation of quadratic terms and linear terms, respectively, while center points offer an internal estimation of pure error. Four factors like sliding velocity, normal load, sliding distance, and fiber content are considered as independent variables and wear rate as a response.^{21,22} Factors and their stages are given in Table 3.

Artificial neural network

The concept of artificial neural networks is used in various engineering applications to solve different complex problems. Artificial neural networks are computing structures inspired by the genetic neural networks of the human brain. This network contains artificial neurons and its process information, which is used for the creation of a systematic model and optimization of the physical and nonphysical complex system. ANN works on the three fundamental principles, specifically (i) validation, (ii) training, and (iii) testing. The whole operation of ANN is performed in three layers, input layer, hidden layer, and output layer, which involves database training to estimation of input/output correlation. The data is given to the input layer, processed in the hidden layer, and finally, the outcomes are exported in the output layer. The detailed procedure of ANN is described by Kadi.²³ In the present work, ANN is used to predict the specific wear rates in different combinations of operating parameters.

Study of worn surfaces

A scanning electron microscope (JEOL JSM 6480LV) is used to obtain the micrographs of worn surfaces of epoxy-SBF composite. Before taking the images, the worn-out surface of test samples is coated with platinum with the help of a sputter coating machine (JEOL JFC-1600) to enhance the conductivity of the worn surface.

Results and discussion

Tensile strength

The tensile strength of composite samples with various proportions of fiber content (0, 2, 4, 6, and 8 wt. %) are measured and the results of the tests are graphically shown in Figure 1. It can be observed that up to 6 wt.% of fiber content the tensile

Table 3. Control factors and their stages for sliding wear.

Control factors	Stages			Symbol
	Low (−I)	Mid (0)	High (+I)	
Sliding velocity (cm/s)	50	125	200	A
Normal load (N)	5	12.5	20	B
Sliding distance (m)	200	500	800	C
Fiber content (wt.%)	0	3	6	D

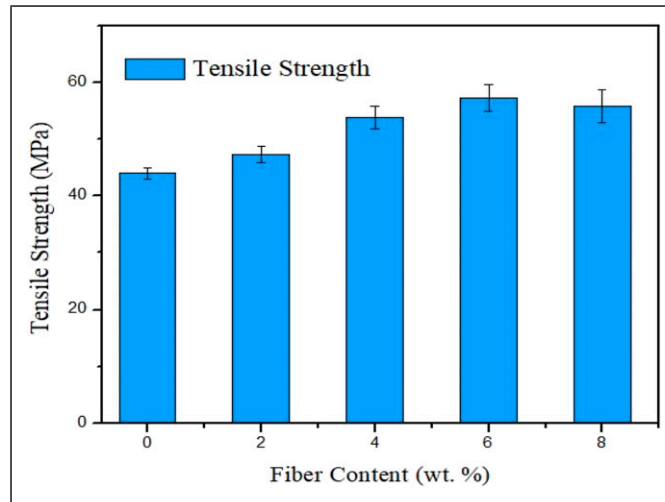


Figure 1. Tensile strength of epoxy-SBF composites at different fiber loadings.

strength of the composite shows an increasing trend. The increase in tensile strength of the composite may be due to the transfer of the major portion of the load from matrix to the fiber during tensile loading. However, further addition of fiber, decreases the tensile strength of the composite. The possible reason for this is that with the increase in fiber concentration weak interfacial bonding may occur due to the insufficient resin and the load was unable to transfer properly from matrix to fiber. A similar observation has been reported by Pokhriyal et al. and Rahman et al.^{24,25}

Flexural strength

Figure 2 shows the flexural strength of the epoxy-SBF composites at different fiber loadings. A monotonic increment in the flexural strength can be observed with the increase in fiber content in the composite. The reason may be because of the proper fiber-matrix interaction under the transverse loading during the test.²⁵

Specific wear rate

The specific wear rates for all 30 experiments are presented in Table 4. The experiments are performed by setting the control factors at selected levels according to the face-centered central composite design of response surface methodology (RSM). It can be observed that with the incorporation of betel nut fiber the wear resistance characteristic of the composite is improved. This maybe because of the proper interfacial bonding between the fiber and epoxy matrix due to the presence of trichomes on the fiber surface.

Analysis of RSM test results

In the present research, the quadratic model is found to be statistically appropriate for the analysis of sliding wear rate of epoxy-SBF composites by response surface methodology. The Analysis of Variance (ANOVA) table obtained from this model is presented in Table 5. It is observed that the terms A (Sliding Velocity), B (Normal Load), D (Fiber Content), AD (Sliding Velocity–Fiber Content), and D² (Fiber Content—Fiber Content) are significant due to their probability factor is less than 0.10 ($p < .10$). The values of R², adjusted R², and predicted R² are 0.997, 0.994, and 0.988, respectively, which indicates that the relationship between response and independent process parameters is outstanding. Adeq Precision estimates signal to noise proportion. A proportion more than 4 is desired. Here, precision value is 56.092 which suggests that the developed model is good enough for the prediction of specific wear rate. The value of the lack of fit of the model is above 0.05, which shows that it is insignificant in nature. An insignificant lack of fit indicates that the terms left out from the model are not significant.

The predictive equation with regression coefficients obtained from the quadratic model of sliding wear rate analysis of epoxy-SBF composites is presented in

$$\begin{aligned}
 W_s = & (11.47) + (1.42*A) - (0.2717*B) - (0.1789*C) - (7.46*D) + (0.0219*AB) + \\
 & (0.0344*AC) - (0.6656*AD) + (0.0519*BD) + (0.0444*CD) + (0.4221*A^2) - \\
 & (0.2729*B^2) - (0.3579*C^2) + (3.26*D^2)
 \end{aligned} \quad (3)$$

Figure 3 shows the normal probability plot for the residuals of the sliding wear rate. Residuals are the difference between experimental and predicted values. It can be noticed that the residuals are laid almost in a linear line, which shows that errors are normally distributed.

Figure 4 shows the experimental and predicted specific wear rates of epoxy-SBF composite. It can be noticed that the results obtained from the proposed correlations are in good agreement with the experimental values.

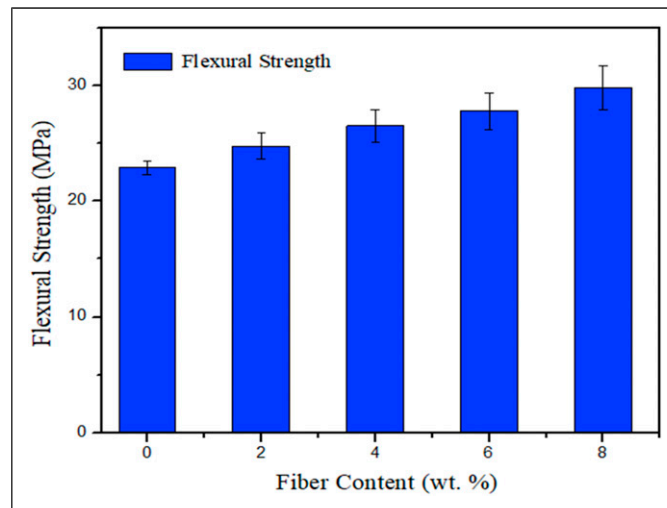


Figure 2. Flexural strength of epoxy-SBF composites at different fiber loadings.

Table 4. Specific wear rate of epoxy-SBF composite samples.

Test run	Sliding Velocity (cm/s) A	Normal Load (N) B	Sliding Distance (m) C	Fiber Content (wt. %) D	Sp. Wear Rate ($\times 10^{-3}$ mm ³ /N-m)
1	125	12.5	500	3	10.97
2	200	5	200	0	24.12
3	125	12.5	500	3	11.18
4	125	12.5	500	6	7.06
5	125	12.5	500	3	11.32
6	50	5	800	0	19.78
7	125	12.5	500	3	11.37
8	200	12.5	500	3	13.76
9	125	12.5	500	0	22.75
10	125	12.5	800	3	10.53
11	50	20	200	0	19.64
12	125	12.5	200	3	12.04
13	125	12.5	500	3	11.15
14	50	5	200	0	20.54
15	200	20	200	0	23.95
16	200	5	800	6	7.92
17	200	20	200	6	7.62
18	200	5	800	0	24.02
19	125	5	500	3	12.32
20	200	20	800	6	7.57
21	50	12.5	500	3	10.37
22	125	20	500	3	10.42
23	50	20	800	0	19.46
24	200	20	800	0	23.78
25	50	5	200	6	6.56
26	125	12.5	500	3	11.82
27	200	5	200	6	8.18
28	50	20	800	6	6.24
29	50	5	800	6	6.41
30	50	20	200	6	6.28

Effect of control factors on the wear rate

Figure 5(a) presents the impact of sliding velocity and fiber content on the specific wear rate of the composite. It is observed that with the increase in sliding velocity, the specific wear rate increases. This maybe because of the deterioration of mechanical properties of the composite as well as interfacial adhesion between the fiber and the matrix due to the generation of heat at the contact surface of the two materials with the increase in sliding velocity.²⁶

Table 5. ANOVA table for sliding wear rate of epoxy-SBF composite.

Source	Sum of squares	DOF	Mean square	F-value	p-value	
Model	1116.683	14	79.763	360.325	<.0001	Significant
A-velocity	36.523	1	36.522	164.989	<.0001	
B-load	1.328	1	1.328	6.002	.0271	
C-distance	0.576	1	0.576	2.602	.1275	
D-fiber	1000.535	1	1000.535	4519.863	<.0001	
AB	0.008	1	0.007	0.034	.8550	
AC	0.019	1	0.018	0.085	.7741	
AD	7.089	1	7.088	32.023	<.0001	
BC	0.043	1	0.043	0.194	.6655	
BD	0.004	1	0.004	0.020	.8878	
CD	0.032	1	0.031	0.142	.7113	
A ²	0.463	1	0.461	2.085	.1693	
B ²	0.193	1	0.193	0.871	.3653	
C ²	0.332	1	0.332	1.499	.2397	
D ²	27.571	1	27.571	124.549	<.0001	
Residual	3.320	15	0.221			
Lack of fit	2.898	10	0.289	3.439	.0926	not significant
Pure error	0.421	5	0.084			
Cor total	1120.00	29				
Std. Dev	0.470			R ²	.997	
Mean	13.304			Adj. R ²	.994	
C.V. %	3.536			Pred. R ²	.988	
				Adeq precision	56.092	

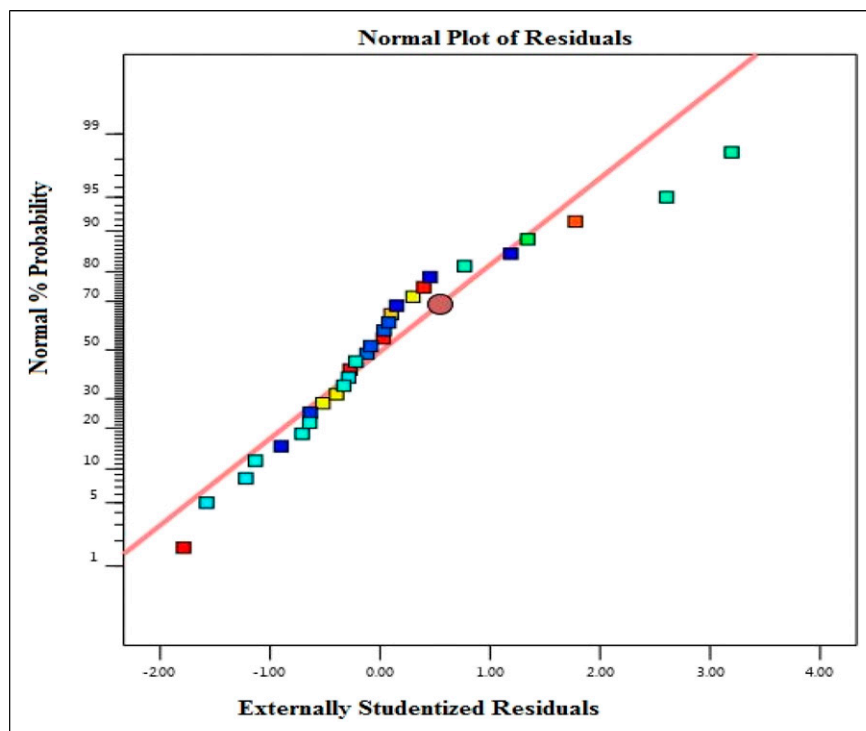
**Figure 3.** The normal plot of residuals for epoxy-SBF composite.

Figure 5(b) shows the impact of normal load and fiber content on wear rate. It can be noticed that with the increase in wt.% of the fiber, the specific wear rate decreased. The reason behind this is, the reinforced fibers act as a barrier in the softening of the material surface due to the rising frictional heat during sliding.²⁷ This may also be due to the increase in hardness as well as the compressive strength of the composite due to the reinforcement of fiber in the epoxy matrix.^{24,27}

Figure 5(c) shows the impact of sliding distance and fiber content on the wear performance of the composite. It can be seen that sliding distance has not much effect on the wear rate of the composite as compared to other factors.

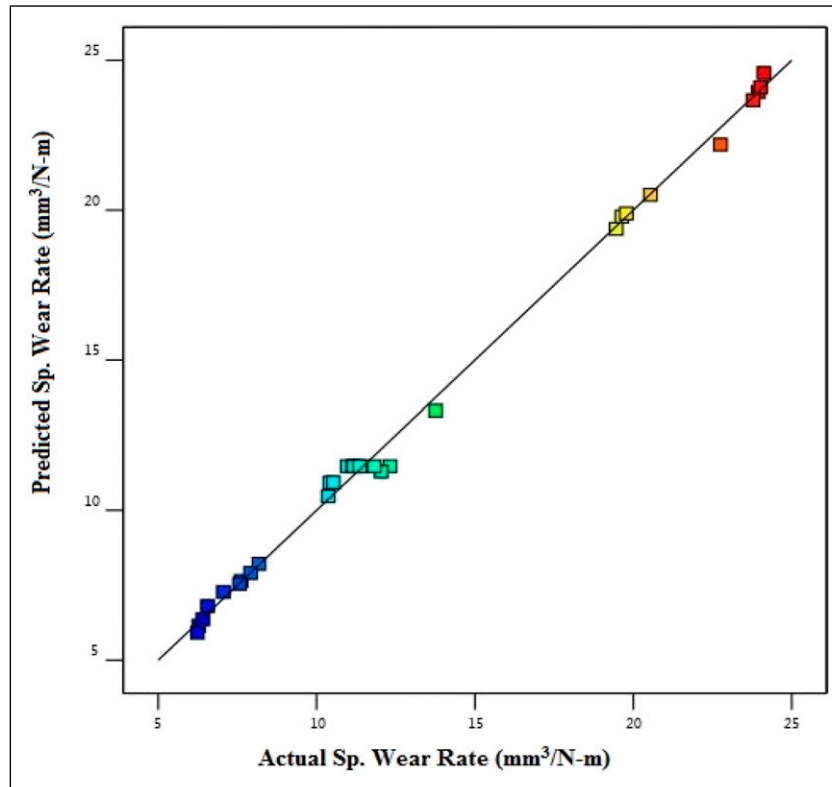


Figure 4. Predicted vs Actual specific wear rate of epoxy-SBF composite.

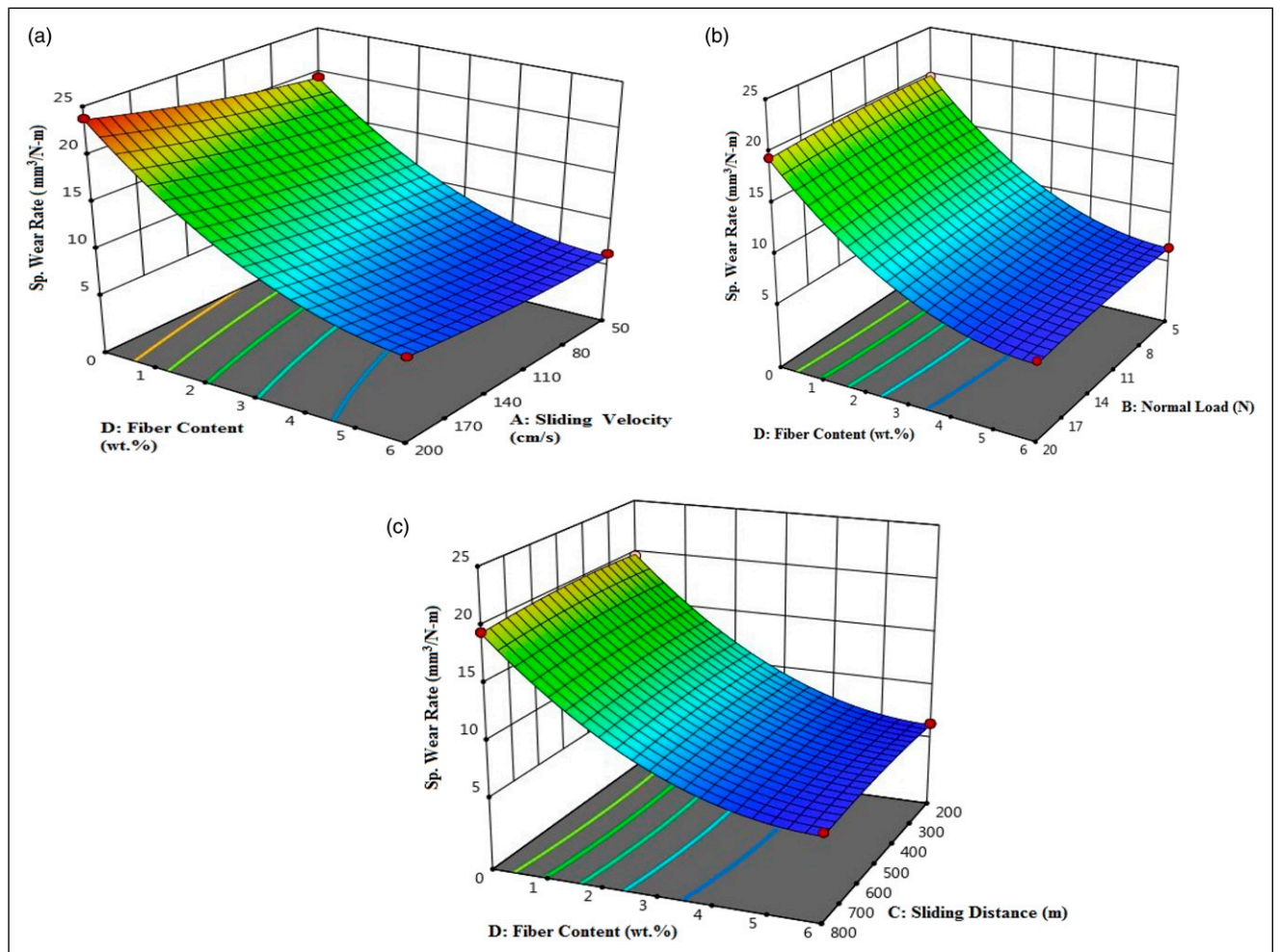


Figure 5. (a) Effect of sliding velocity and filler content on specific wear rate of epoxy-SBF composite. (b) Effect of fiber content and normal load on specific wear rate of epoxy-SBF composite. (c) Effect of fiber content and sliding distance on specific wear rate of epoxy-SBF composite.

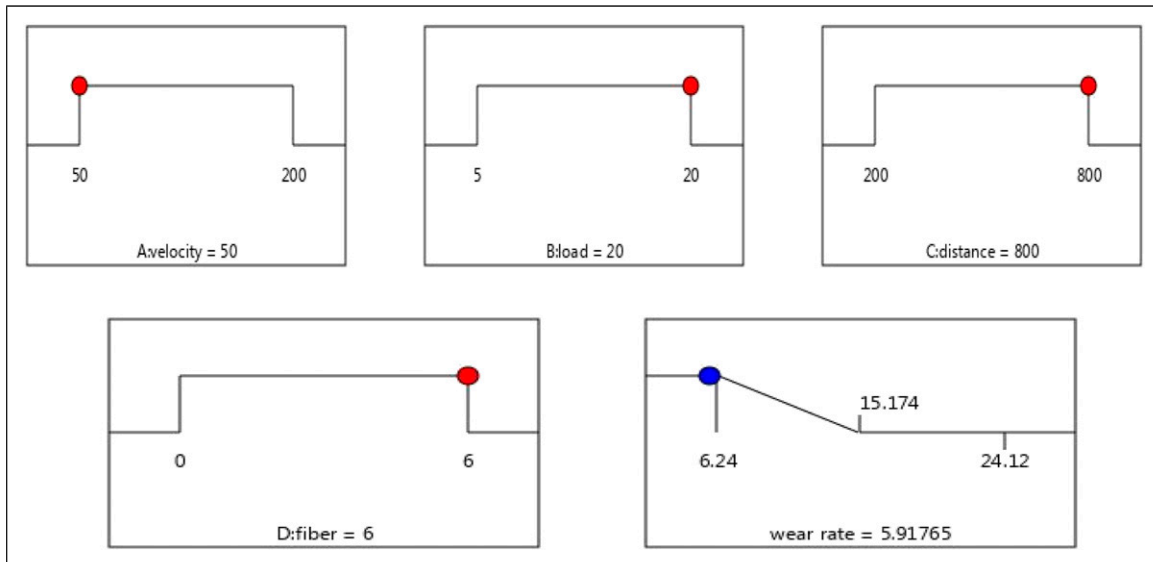


Figure 6. Optimal conditions for the least specific wear rate.

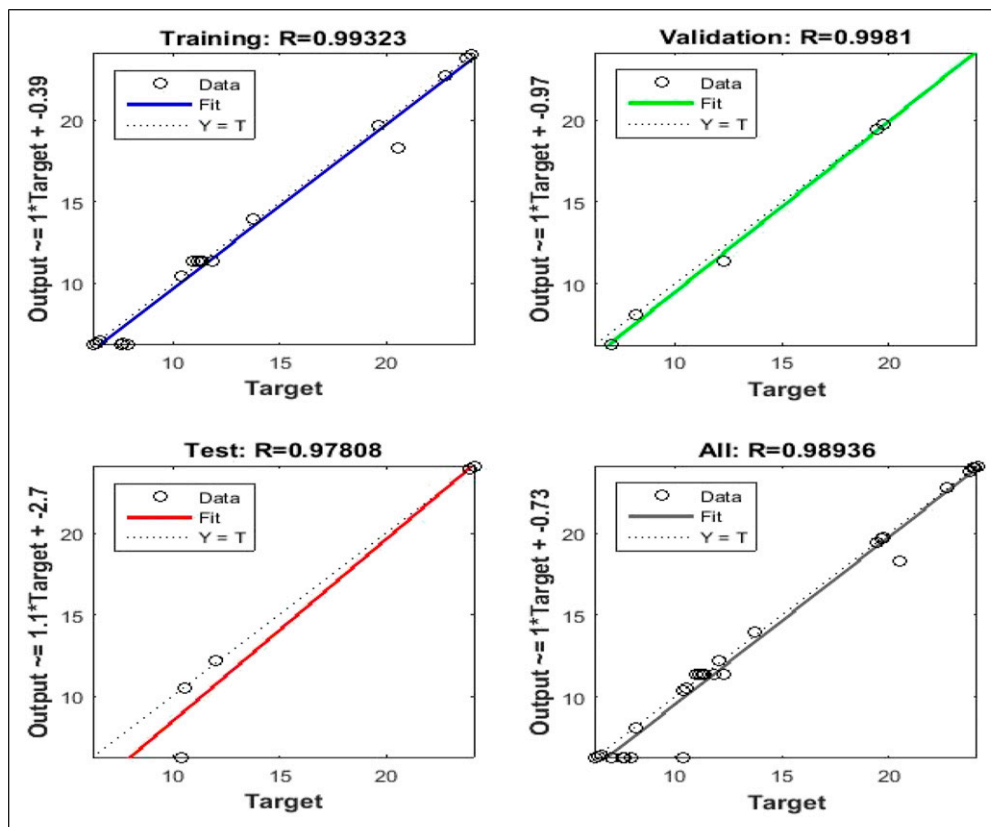


Figure 7. Performance plots for sliding wear rate prediction of epoxy-SBF composite.

Prediction of optimal conditions

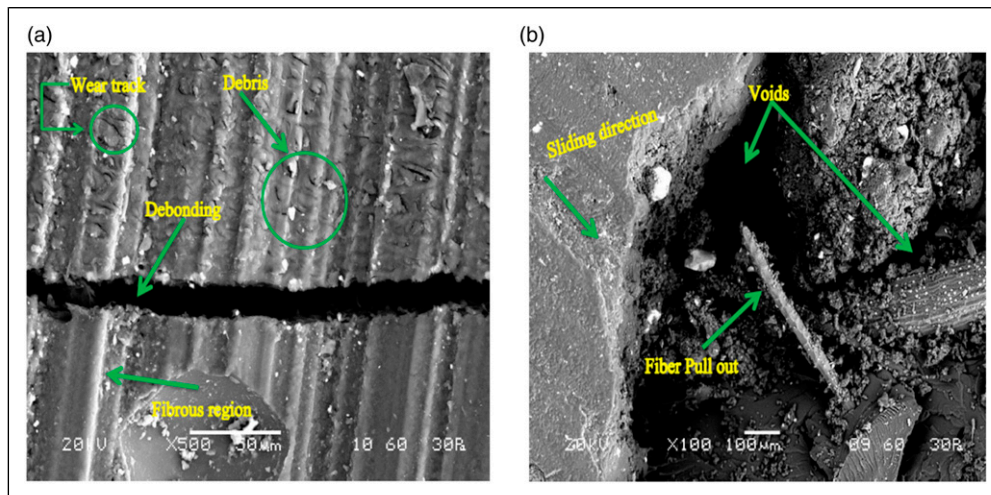
Figure 6 presents the optimal condition at which the wear rate is minimum. The optimal set of control factors for the least specific wear rate ($5.92 \times 10^{-3} \text{ mm}^3/\text{Nm}$) is found to be sliding velocity 50 cm/s, normal load 20 N, sliding distance 800 m, and fiber content 6.5 wt.%, where the desirability is one.

Prediction using artificial neural network

To predict the wear rate of the composite samples at various parametric conditions, a neural network based on the back-propagation ANN method has been developed using the existing experimental database. A neural network with the least error has been selected for training with the exploration of the number of networks by varying the values of input training

Table 6. Comparison of experimental specific wear rate with RSM and ANN predicted results.

Test run	Sp. Wear Rate (10^{-3} mm ³ /N-m) (Experimental)	Sp. Wear Rate (10^{-3} mm ³ /N-m) (RSM Predicted)	Error (%)	Sp. Wear Rate (10^{-3} mm ³ /N-m) (ANN Predicted)	Error (%)
1	10.97	11.47	4.576	11.35	3.464
2	24.12	24.57	1.898	24.12	0
3	11.18	11.47	2.612	11.35	1.521
4	7.06	7.27	3.088	6.24	11.615
5	11.32	11.47	1.343	11.35	0.2655
6	19.78	19.89	0.566	19.77	0.051
7	11.37	11.47	0.897	11.35	0.176
8	13.76	13.32	3.205	13.97	1.526
9	22.75	22.19	2.461	22.76	0.044
10	10.53	10.93	3.846	10.54	0.095
11	19.64	19.78	0.743	19.70	0.305
12	12.04	11.29	6.196	12.19	1.245
13	11.15	11.47	2.887	11.35	1.794
14	20.54	20.51	0.136	18.31	10.856
15	23.95	23.94	0.033	23.99	0.167
16	7.92	7.91	0.126	7.08	10.606
17	7.62	7.64	0.315	6.89	9.521
18	24.02	24.10	0.321	24.02	0
19	12.32	11.47	6.883	11.36	7.792
20	7.57	7.54	0.291	6.99	7.568
21	10.37	10.47	0.955	10.41	0.385
22	10.42	10.92	4.866	9.529	8.542
23	19.46	19.37	0.432	19.45	0.051
24	23.78	23.66	0.471	23.78	0
25	6.56	6.81	3.780	6.46	1.524
26	11.82	11.47	2.936	11.35	3.976
27	8.18	8.21	0.416	8.10	0.977
28	6.24	5.91	5.1606	6.24	0
29	6.41	6.36	0.671	6.32	1.404
30	6.28	6.15	2.038	6.24	0.636

**Figure 8.** SEM images of worn samples of epoxy-SBF composites (a) fiber content 6 wt.% and sliding velocity 200 cm/s (b) fiber content 3 wt.% and sliding velocity of 125 cm/s.

parameters. To perform the ANN analysis, the values of the input parameters are normalized to lie in the range of 0–1. Figure 7 shows a regression plot of training, validation, testing, and all data sets. From the graph, it can reveal that the overall regression coefficient (R^2) is 0.9894, which shows the developed neural network is appropriate for the prediction of the wear rate of the composite samples.

The comparison between experimentally obtained results with RSM and ANN predicted results are given in Table 6. It can be noticed that the predicted wear rate values obtained from ANN analysis are in good agreement with the experimental results compared to RSM predicted results.

Morphological study of the worn sample

The morphologies of worn samples are shown in Figure 8. Figure 8(a) is the micrograph of a composite sample with 6 wt% fiber content after sliding over a distance of 200 m, at a speed of 200 cm/s and under an applied load of 20 N. Wear track, debonding, and formation of debris are clearly visible from this micrograph. Due to contact of samples with the rotating disc, development of wear track gradually begins. During the tribological test, fibers bear the maximum amount of applied load. Hence, as fiber wt% increases in the composite, debonding may occur in the fibrous region. Figure 8(b) represents SEM image of 3 wt% of fiber content tested at 500 m sliding distance, sliding velocity of 125 cm/s, and under a load of 12.5 N. Sliding direction is mentioned through wear track which indicates the plastic flow of material. Material removal due to plowing action as a result of repeated sliding can also be evident from this micrograph.

Conclusions

1. Successfully fabrication of epoxy-SBF composites at different wt.% of fiber contents is possible through simple hand lay-up route.
2. Mechanical properties of epoxy can be modified with reinforcement of betel nut fiber.
3. Using the RSM, a quadratic model is developed. From the ANOVA table, it can be concluded that fiber content and sliding velocity are major influencing factors affecting the specific wear rate of the composites.
4. ANN is used to predict the specific wear rates of epoxy-SBF composites. The predicted values of wear rates exhibited good agreement with experimental values which confirms the remarkable capability of the neural network used in the present study.

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ORCID iDs

Purushottam Kumar Choudhary  <https://orcid.org/0000-0003-3670-1530>

Bishnu Prasad Nanda  <https://orcid.org/0000-0001-9638-9628>

References

1. Holbery J and Houston D. Natural-fiber-reinforced polymer composites in automotive applications. *Jom* 2006; 58: 80–86.
2. Pickering KL, Efendy MGA and Le TM. A review of recent developments in natural fibre composites and their mechanical performance. *Compos A Appl Sci Manuf* 2016; 83: 98–112.
3. Thakur VK, Singha AS and Thakur MK. Ecofriendly biocomposites from natural fibers: mechanical and weathering study. *Int J Polym Anal Charact* 2013; 18: 64–72.
4. John MJ and Thomas S. Biofibres and biocomposites. *Carbohydr Polym* 2008; 71: 343–364.
5. Choudhary PK, Pradhan P and Satapathy A. Study on mechanical and tribological characteristics of epoxy composites reinforced with short areca-catechu fibers. *J Nat Fibers* 2021; 1: 13.
6. Shrivastava R, Telang A, Rana RS, et al. Mechanical properties of coir/G lass fiber epoxy resin hybrid composite. *Mater Today Proc* 2017; 4: 3477–3483.
7. Jacob M, Thomas S and Varughese KT. Mechanical properties of sisal/oil palm hybrid fiber reinforced natural rubber composites. *Compos Sci Technol* 2004; 64: 955–965.
8. Pothan LA, Oommen Z and Thomas S. Dynamic mechanical analysis of banana fiber reinforced polyester composites. *Compos Sci Technol* 2003; 63: 283–293.
9. Mohan K and Rajmohan T. Tribological characteristics of natural fiber composite - Review. *IOP Conf Ser Mater Sci Eng* 2020; 954: 012050.
10. Yallem TB, Kumar P and Singh I. Sliding behaviour of woven industrial hemp fabric reinforced thermoplastic polymer composites. *Int J Plast Technol* 2015; 19: 347–362.
11. Patnaik A, Satapathy A, Dwivedy M, et al. Wear behavior of plant fiber (Pine-Bark) and cement kiln dust-reinforced polyester composites using Taguchi experimental model. *J Compos Mater* 2010; 44: 559–574.
12. Sabeel Ahmed K, Khalid SS, Mallinatha V, et al. Dry sliding wear behavior of SiC/Al₂O₃ filled jute/epoxy composites. *Mater Des* 2012; 36: 306–315.

13. Nirmal U, Yousif BF, Rilling D, et al. Effect of betelnut fibres treatment and contact conditions on adhesive wear and frictional performance of polyester composites. *Wear* 2010; 268: 1354–1370.
14. Yousif BF, Lau STW and McWilliam S. Polyester composite based on betelnut fibre for tribological applications. *Tribol Int* 2010; 43: 503–511.
15. Ashok RB, Srinivasa CV and Basavaraju B. A review on the mechanical properties of areca fiber reinforced composites. *Sci Technol Mater* 2018; 30: 120–130. DOI: [10.1016/j.stmat.2018.05.004](https://doi.org/10.1016/j.stmat.2018.05.004).
16. Lazim Y, Salit SM, Zainudin ES, et al. Effect of alkali treatment on the physical, mechanical, and morphological properties of waste betel nut (Areca catechu) husk fibre. *BioResources* 2014; 9: 7721–7736.
17. Hassan MM, Wagner MH and Hayder U. Study on the performance of hybrid jute/betel nut fiber reinforced polypropylene composites. *J Adhes Sci Technol* 2012; 25: 615–626.
18. Rahman MM, Mondol M and Hasan M. Mechanical properties of chemically treated coir and betel nut fiber reinforced hybrid polypropylene composites. *IOP Conf Ser Mater Sci Eng* 2018; 438: 012025. DOI: [10.1088/1757-899X/438/1/012025](https://doi.org/10.1088/1757-899X/438/1/012025).
19. Yusriah L, Sapuan SM, Zainudin ES, et al. Exploring the potential of betel nut husk fiber as reinforcement in polymer composites: effect of fiber maturity. *Proced Chem* 2012; 4: 87–94.
20. Nanda BP and Satapathy A. Wear characteristics of epoxy based hybrid composites reinforced with short hair fibers and glass micro spheres. *Materials Today: Proceedings* 2018; 5: 12588–12594. DOI: [10.1016/j.matpr.2018.02.241](https://doi.org/10.1016/j.matpr.2018.02.241).
21. Nanda BP and Satapathy A. An analysis of the sliding wear characteristics of epoxy-based hybrid composites using response surface method and neural computation. *J Nat Fibers* 2020; 00: 1–15.
22. Nayak A, Kumar Jesthi D, Chandra Routara B, et al. Tribological properties of glass/carbon hybrid composites through inter-ply arrangement using Response Surface Methodology. *Mater Today Proc* 2018; 5: 19828–19835.
23. El Kadi H. Modeling the mechanical behavior of fiber-reinforced polymeric composite materials using artificial neural networks—A review. *Compos Struct* 2006; 73: 1–23.
24. Pokhriyal M, Prasad L, Rakesh PK, et al. Influence of fiber loading on physical and mechanical properties of Himalayan nettle fabric reinforced polyester composite. *Mater Today Proc* 2018; 5: 16973–16982.
25. Rahman MR, Huque MM, Islam MN, et al. Improvement of physico-mechanical properties of jute fiber reinforced polypropylene composites by post-treatment. *Compos Part A Appl Sci Manuf* 2008; 39: 1739–1747.
26. Sudheer M, Prabhu R, Raju K, et al. Modeling and analysis for wear performance in dry sliding of epoxy/glass/ptw composites using full factorial techniques. *ISRN Tribol* 2013; 2013: 1–11.
27. Singh T, Gangil B, Singh B, et al. Natural-synthetic fiber reinforced homogeneous and functionally graded vinylester composites: effect of bagasse-Kevlar hybridization on wear behavior. *J Mater Res Technol* 2019; 8: 5961–5971.

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